

Team Gold: Rogue-Like Algebra

Lab 1: Outline

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1 Introduction

Many students struggle to understand algebra due to traditional teaching methods that fail to engage learners effectively. In the 2020-21 school year, only 74% of over 2 million high schoolers passed their Algebra I class (U.S. Department of Education). This failure leads to frustration, disengagement, and a lack of confidence in math skills. They might attempt to seek out learning tools on their own, only to find that these traditional forms of learning still persist even through modern resources. *Figure 1* displays this dilemma, in which students seeking learning materials may find themselves in an unrewarding loop of learning and still require more assistance when these materials fail to help them.

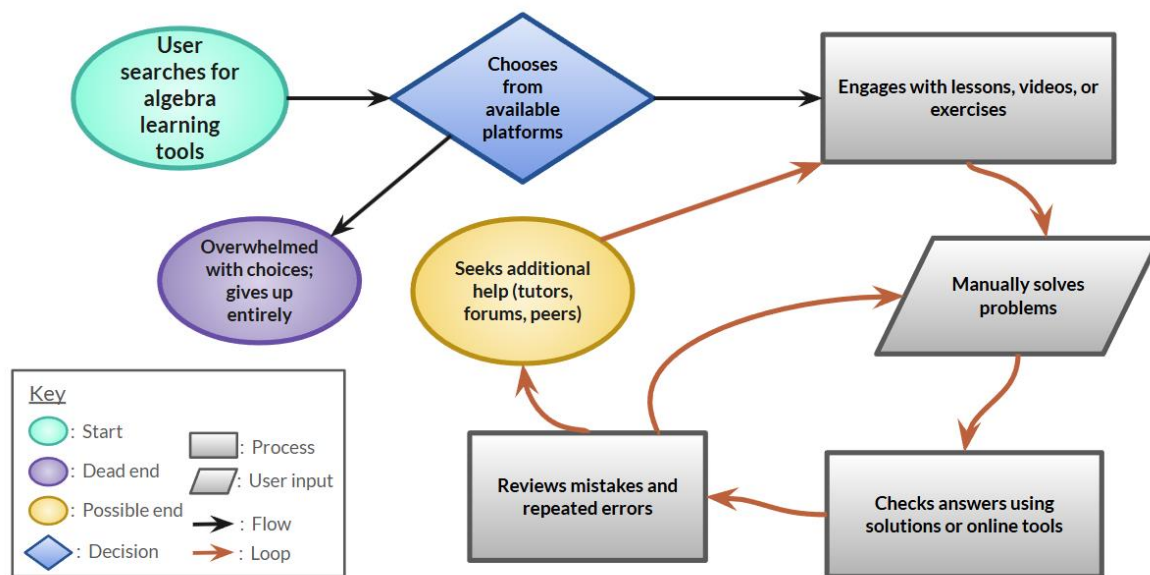


Figure 1: Current Process Flow

A solution must address the need for interactive, engaging, and adaptive learning tools that make algebra approachable and rewarding. The aim is to develop a game that will help those who struggle to understand algebra. *Rogue-Like Algebra* will be a turn-based, rogue-like adventure designed to gamify algebra. With dungeons, bosses, and power-ups all throughout, the player will be as engaged in the world as they are with the math. By integrating strategic gameplay with problem-solving, learning can be transformed into an immersive and fun experience. The game aims to foster confidence in math skills while making algebra fun and engaging for middle and high school students.

2 Rogue-Like Algebra Product Description

Rogue-Like Algebra is a turn-based adventure game that transforms the challenge of learning algebra into an engaging and immersive experience, as shown in *Figure 2*. Designed to address the struggles many face with algebraic concepts, the game blends strategic dungeon-crawling with problem-solving, where players use algebra to attack, defend, and progress. Mistakes carry consequences—such as reduced combat efficiency or character penalties. The game's procedurally generated dungeons and adaptive difficulty ensure replayability and personalized challenge. Aimed primarily at middle and high school students, but accessible to all learners, *Rogue-Like Algebra* provides an educational alternative to traditional rote memorization by offering a gamified system that rewards creativity, risk-taking, and improvement. The game is available on the web and can be played in any modern browser. The project's goal is to foster confidence in math skills while making algebra fun, approachable, and rewarding.

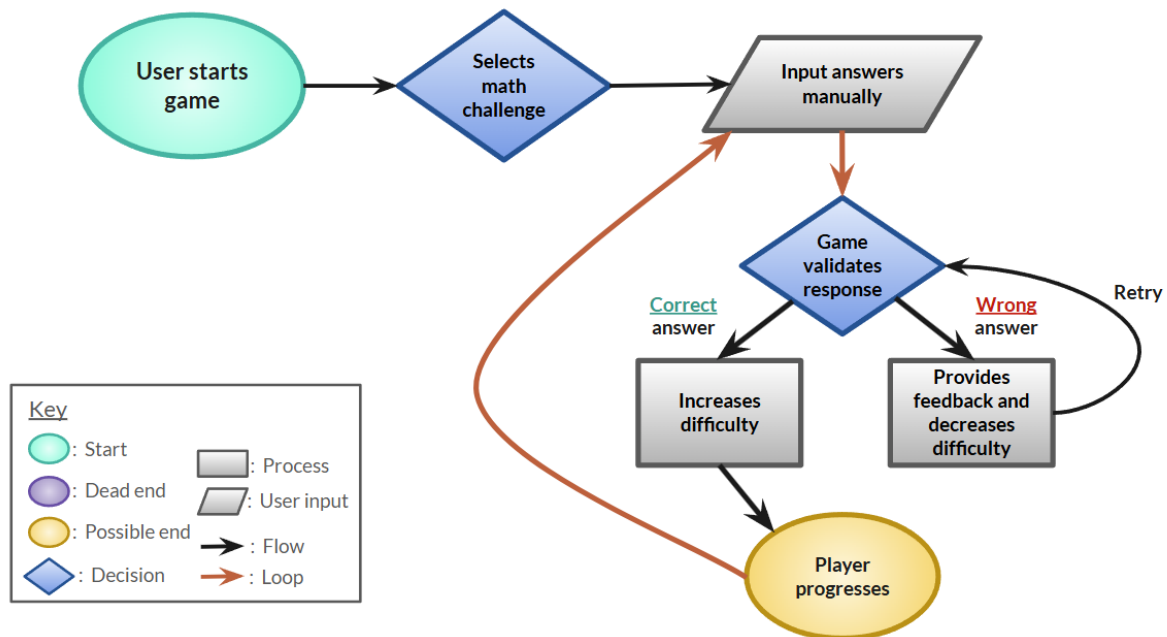


Figure 2: Solution Process Flow

1.1. Key Product Features and Capabilities

Gameplay - Turn-based combat, immersive storyline, and minigames

Engagement - Adaptive systems to scale difficulty, time limits, and boss battles

Objective - Gamify and reduce complexity of learning algebra

How it Solves the Problem

- Rewards players for solving difficult problems and completing achievements
- Removes risk of *rote memorization* through procedural generation and multiple accepted approaches to solving problems

1.2. Major Components

1.2.1. IDE - VS Code

- 1.2.1.1. Powerful, extensible, free.

1.2.2. Version Control - Git/GitHub

- 1.2.2.1. Industry standard, suggested by investors (professor).

1.2.3. CI/CD - GitHub Actions & Workflows

- 1.2.3.1. Integrated with GitHub

1.2.4. Target OS - Windows & Linux

- 1.2.4.1. Windows is the most popular end-user platform.
- 1.2.4.2. Linux is the backbone of mobile devices.

1.2.5. Engine - Godot

- 1.2.5.1. Free & open source.
- 1.2.5.2. Extensible.
- 1.2.5.3. A lot of free tutorials and documentation.

1.2.6. Language - Rust & GDScript

- 1.2.6.1. Rust is fast, modern, and safe.
- 1.2.6.2. GDScript must be used for edge cases which aren't covered by Rust.

1.2.7. Systems

- 1.2.7.1. Character controllers must be created or adapted from freely available templates.
- 1.2.7.2. A Leaderboard system must be created.

- 1.2.7.3. Procedural generators must be developed for dungeons and math problems.
- 1.2.7.4. Enemy AI must be created and tuned.

3 Identification of Case Study

Rogue-Like Algebra is primarily for middle and high school students who will likely take an algebra course or an otherwise related math. STEM-oriented players and puzzle game enthusiasts will find the game tailored to their interests as well, making it both educational for new learners and entertaining for those already familiar with algebra. Though these are the groups of people most catered to for the game, other kinds of players can gain just as much value from it. Educators, the parents of learning students, casual and indie genre fans, and those who enjoy logic and strategy-based games can also benefit from the purpose of *Rogue-Like Algebra*.

The primary purpose of the game is to assist in the learning and reinforcement of foundational algebraic principles of those who play it in the most efficient and effective way possible. The game can provide the player with both a satisfying and fun experience as to retain attention in the learning process and prevent burnout or frustration. It can also help reinforce and refresh algebraic concepts of those who aren't the primary audience who might need a review before moving on to higher math. Overall, *Rogue-Like Algebra* contains an enjoyable and repeatable game loop experience to all audiences.

4 Glossary

1.3. Terms

- 1.3.1. **Adaptive Learning** – An educational method that adjusts content difficulty based on the learner’s performance to personalize the learning experience.
- 1.3.2. **Algebraic Concepts** – Foundational mathematical principles involving variables, equations, expressions, and functions typically taught in middle and high school.
- 1.3.3. **Continuous Integration (CI)** – A development practice where developers frequently merge code changes into a shared repository, triggering automated builds and tests to detect issues early.
- 1.3.4. **Continuous Deployment (CD)** – A software release process that automatically delivers validated code changes to production, minimizing manual intervention and ensuring faster updates.
- 1.3.5. **GDScript** – A Python-like scripting language used in the Godot game engine, designed for creating game logic.
- 1.3.6. **Git** – A distributed version control system that tracks changes in source code during software development, enabling collaboration and history tracking.
- 1.3.7. **GitHub** – A cloud-based platform for hosting Git repositories. It offers tools for collaboration, issue tracking, code review, and CI/CD integration.
- 1.3.8. **Godot** – A free and open-source game engine used for developing 2D and 3D games, known for its flexibility and ease of use.
- 1.3.9. **IDE (Integrated Development Environment)** – A software suite that provides tools like a code editor, debugger, and compiler to streamline software development (e.g., VS Code).
- 1.3.10. **Procedural Generation** – A method of creating data algorithmically rather than manually, often used for generating levels, puzzles, or content dynamically in games.
- 1.3.11. **Rogue-like** – A subgenre of games typically featuring procedural generation, turn-based movement, and permanent death, emphasizing strategy and replayability.

- 1.3.12. **Rust** – A systems programming language known for performance, safety, and concurrency.
- 1.3.13. **Turn-Based Combat** – A gameplay mechanic where players and opponents take alternating turns to perform actions such as attacking or defending.
- 1.3.14. **VS Code (Visual Studio Code)** – A popular, lightweight code editor developed by Microsoft, used for writing and debugging code.

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